OPERATIONS & MAINTENANCE MANUAL

Dredging Equipment — All Series

Marine Engineering Division

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1. Introduction & Scope

This Operations and Maintenance Manual provides comprehensive guidance for the safe operation and routine maintenance of four dredger models operated by Benamdaj Dredging Solutions Ltd. It covers pre-start procedures, startup and shutdown sequences, daily and periodic maintenance schedules, operating guidelines, and emergency procedures.

This manual applies equally to all four vessels in the documented series: the 16/14" Cutter Suction Dredger, the 14/12" Amphibious Multifunctional Dredger, the 12/10" Bucket Chain Dredger, and the 10/10" Jet Suction Dredger. Where procedures differ between models this is clearly noted. All dredgers share the same electrical system (CAT 3412 auxiliary), the same 240 V manual control configuration, and the same winch-based positioning system.

WARNING: This manual must be read in full by all operators before commencing dredging operations. Non-compliance with procedures may result in injury, equipment damage, or environmental incidents.

2. Pre-Start Safety Checklist

The following checklist must be completed every morning before the engine is started and before commencing any dredging shift. All items must be confirmed satisfactory before the start sequence is initiated.

> *NOTE: If any item on this checklist cannot be confirmed satisfactory, do not start the engine. Report the issue to the supervisor before proceeding.*

2.1 Daily Pre-Start Checklist

1	Check engine oil level using the dipstick. Level must be between the MIN and MAX marks. Top up with Total Ruby X if below MIN.
2	Check fuel level in diesel tank. Plan refuelling before the shift if level is below half. Refuel only with the engine stopped.
3	Inspect all hose connections, pipe joints, and suction line fittings for signs of leakage. Tighten or replace as needed before starting.
4	Inspect the jet suction head (Models 1, 2) or cutter/suction assembly (Models 3, 4) for debris blockage. Clear any obstruction before lowering to dredging depth.
5	Check all safety guards and protective covers are in place and secured.
6	Confirm anchor lines and mooring ropes are correctly secured to hold vessel in position.
7	Test the emergency stop button to confirm it operates correctly. Reset after test.
8	Confirm a serviceable fire extinguisher is present at the designated location on deck.
9	Confirm all crew members are wearing correct PPE: life jackets, hard hats, safety boots, and high-visibility vests.
10	Check weather and wind conditions. Do not commence dredging if conditions exceed safe operating limits.
11	Check gearbox oil level (Models 1, 2, 3) or PTO condition (Model 4) every morning before startup.
12	Inspect anchor wires and winch lines for signs of kinking, fraying, or wear. Replace before use if damaged.

3. Engine Startup Sequence

Follow these steps in order every time the engine is started. Do not skip steps or vary the sequence.

3.1 Main Engine Startup (all models)

1	Check engine oil level on the gauge/dipstick — confirm it is within the acceptable range before cranking.
2	Check gearbox oil level and condition (Models 1, 2, 3). For Model 4 (PTO), confirm PTO disengaged.
3	Check fuel level. Confirm sufficient fuel for the planned shift duration.
4	Check battery terminals are clean and securely connected. Confirm battery charge is adequate.
5	Set throttle to idle position before engaging the ignition.
6	Turn ignition key to START. Hold until engine fires. Release immediately when engine starts.
7	Allow engine to run at idle for 5 minutes minimum before applying any load. This allows oil to circulate and engine temperature to stabilise.
8	Monitor oil pressure gauge and coolant temperature during warm-up. Confirm both are within normal range before proceeding.
9	Engage auxiliary engine (CAT 3412) if electrical load is required — confirm generator output voltage is 240 V before connecting loads.
10	Confirm no unusual noises, smoke, or warning lights before proceeding to pump engagement.

3.2 Pump & Suction System Engagement

After the main engine has warmed up and all gauges confirm normal operation:

1	Lower the suction ladder/arm to the required dredging depth using the manual winch. Confirm depth is correct before engaging pump.
2	Open suction valve (if fitted) and confirm suction line is clear of obstructions.
3	Prime the pump if required — ensure suction line is submerged before engaging pump drive.
4	Engage pump drive via gearbox (Models 1–3) or PTO (Model 4). Increase engine speed gradually from idle to operating speed.
5	Monitor pump discharge — confirm slurry flow is established within 2 minutes of engagement. If no flow, stop pump and investigate before restarting.
6	Deploy anchor winch lines to establish dredging swing pattern. Confirm all 4 anchors (or applicable number) are set before beginning production swing.
7	Monitor pump pressure gauge — normal operating pressure is 3 bar. Do not exceed 10 bar.
8	Record start time, engine hours, and initial production readings in the operational log.

4. End-of-Shift Shutdown Sequence

WARNING: Never shut down the engine without first flushing the pump. Slurry left in the pump and pipes will solidify and cause blockage or premature wear.

4.1 Shutdown Procedure (all models)

1	Reduce engine speed to idle. Stop dredging swing by securing anchor winches.
2	Open clean water flushing supply to pump suction (or submerge suction in clear water area). Run pump for minimum 5 minutes to flush all slurry from pump casing, impeller, and discharge pipe.
3	Raise suction ladder/arm to the transport/stowed position using the manual winch. Secure with locking pin or rope.
4	Disengage pump drive by disengaging gearbox (Models 1–3) or PTO (Model 4).
5	Reduce engine speed to idle. Allow 3–5 minutes cool-down at idle before shutdown.
6	Turn ignition key to OFF. Confirm engine has fully stopped.
7	Switch off auxiliary engine (CAT 3412). Confirm all electrical loads are disconnected.
8	Secure all anchor wires and mooring lines. Confirm vessel is correctly moored for overnight.
9	Remove and relocate any dredged material (sediments/mud) from the work area as required.
10	Complete the end-of-day operational log — record engine hours, production volume, and any faults or incidents.

4.2 Post-Shutdown Checks

1	Check engine oil level after shutdown and top up if needed — oil level drops slightly when engine is hot.
2	Inspect pump seal/gland for leakage and record condition. Replace seal packing if excessive drip is observed.
3	Inspect all hose connections and pipe joints for leaks that developed during the shift.
4	Check anchor wires and winch drums — inspect for kinks, fraying, or uneven spooling.
5	Confirm fire extinguisher is back in its designated location.
6	Report any defects, unusual noises, or production issues to the supervisor before leaving the vessel.

5. Routine Maintenance Schedules

All maintenance intervals are the same across all four models unless specifically noted. Use Total Ruby X engine oil for all engine oil checks and top-ups.

> *NOTE: Maintenance intervals are specified in operating hours. Where both a time period and an hour count are given, use whichever comes first.*

5.1 Daily Maintenance (every shift / every 8–10 operating hours)

TASK	ACTION & STANDARD
Engine oil level	Check dipstick — must be between MIN and MAX. Top up with Total Ruby X if below MIN. Record reading.
Coolant level	Not applicable to these models — dry cooling configuration. Confirm engine cooling fan is operational.
Fuel level	Check diesel tank level every hour during operation. Plan refuelling before tank drops below quarter full.
Hoses, pipes & connections	Visual inspection every morning. Check for cracks, abrasion, loose clamps, and seepage. Tighten or replace immediately.
Pump seal / gland	Check for leaks at all times during operation. A slow drip (2–3 drops/min) is acceptable; steady flow requires immediate attention.
Gearbox oil level & temperature	Check every morning before startup (Models 1, 2, 3). Confirm oil is at correct level and no leaks are present.
Engine noise & vibration	Monitor every hour. Stop engine immediately if unusual knocking, rattling, or vibration is detected.
Anchor wires / winch lines	Inspect for wear, kinking, and fraying every morning. Replace any wire showing broken strands.
Jet suction / cutter assembly	Inspect for debris and wear every morning. Clear any blockage before commencing operations.
Safety equipment	Confirm fire extinguisher, life rings, and life jackets are present and accessible at all times.

5.2 Weekly Maintenance (every 7 days or 50 operating hours)

TASK	ACTION & STANDARD
Engine drive belts	Check tension and condition. Replace if cracked, glazed, or excessively slack. Inspect every day in dusty/sandy conditions.
Engine air filter	Inspect and clean. Replace if clogged — a blocked air filter causes black smoke and power loss.
Fuel filter / water separator	Inspect and drain water separator. Replace fuel filter element if flow is restricted.
Battery terminals	Check for corrosion and security. Clean with baking soda solution if corroded. Confirm charge level.
Ladder pivot points	Grease all pivot bearings every 3 days in normal conditions, or daily in high-sand/abrasive environments.

TASK	ACTION & STANDARD
Spud / anchor pins and cables	Inspect all anchor pins, shackles, and cable connections for wear, deformation, or corrosion. Replace worn pins.

5.3 Service at 200 Hours (Monthly equivalent)

TASK	ACTION & STANDARD
Engine oil and filter	Drain and replace engine oil. Replace oil filter. Use Total Ruby X — confirm correct grade with engine manual. Record date and hours.
Fuel filters	Replace primary and secondary fuel filter elements. Bleed fuel system after replacement.
Hydraulic oil	Sample and test hydraulic oil. Change if contaminated or at 200-hour interval. Check hydraulic system for leaks.
Gearbox oil	Drain and replace gearbox oil (Models 1, 2, 3). Inspect gearbox for leaks and unusual wear. Confirm oil level after fill.
Pump wear plates	Measure wear plate thickness. Replace if below minimum thickness (confirm minimum with pump manufacturer).
Cutter / suction head	Full inspection — replace worn teeth on cutter models. Inspect suction head for erosion and cracking.
Winch drums and cables	Full inspection of all winch drums, cables, and braking mechanisms. Replace any cable showing more than 10% broken strands.
All bolts and fasteners	Check torque on all critical fasteners — pump mounting, engine mounts, ladder pivots, anchor fairleads.
Electrical connections	Inspect all electrical connections for corrosion, loose terminals, and damaged insulation.

5.4 Quarterly Service (every 90 days)

TASK	ACTION & STANDARD
Gearbox full inspection	Drain gearbox, inspect gear teeth and bearings for wear, refill with correct grade oil (Models 1, 2, 3).
Cutter head full rebuild	Disassemble cutter head, inspect shaft bearings, replace worn teeth and seals. Reassemble and grease.
Hull survey	Dry-dock or haul out for underwater hull inspection. Check for corrosion, erosion, and structural damage.
Winch system overhaul	Inspect winch drums, cables, brakes, and fairleads. Replace any worn components.
All safety systems	Full certified inspection of life-saving appliances, fire extinguishers, and emergency stop systems.

5.5 Annual Service (every 12 months or 1,000 hours)

TASK	ACTION & STANDARD
Engine major service	Full service per Weichai / Caterpillar manufacturer schedule. All filters, belts, coolant, and injectors.
Hydraulic fluid full change	Drain entire hydraulic system and refill with fresh fluid. Replace all hydraulic filters.
Pump full overhaul	Disassemble dredge pump completely. Replace impeller, wear plates, seals, and bearings. Inspect casing.
Electrical system inspection	Full inspection and testing of all wiring, panels, generators, and safety circuits by qualified electrician.

6. Operating Guidelines & Limits

6.1 Engine Operating Parameters

TASK	PARAMETER / LIMIT
Normal operating speed	1,200 – 2,100 RPM depending on load and soil conditions
Maximum engine speed — hard limit	1,800 RPM (Weichai models). Never exceed — engine damage will result.
Maximum coolant temperature — alarm	105°C. Stop engine and investigate if temperature reaches or exceeds this.
Normal pump operating pressure	3 bar — monitor continuously during dredging.
Maximum pump pressure — hard limit	10 bar. If exceeded, reduce engine speed and investigate immediately.
Engine warm-up period	Minimum 5 minutes at idle before applying pump or PTO load.
Engine cool-down before shutdown	Minimum 3–5 minutes at idle after disengaging pump.

6.2 Dredging Operating Guidelines by Soil Type

All models use jet suction or centrifugal pump dredging — the following guidelines apply:

TASK	SOIL TYPE / SETTING
Soft silt and mud	Jet suction at full pressure. Maximum swing speed. Monitor discharge for dilute slurry — reduce swing if density is too low.
Fine sand	Jet suction at full pressure with moderate swing speed. Monitor pump pressure — increase engine RPM if pressure drops below 3 bar.
Coarse sand and gravel	Jet suction at maximum pressure. Reduce swing speed to prevent suction head blockage. Monitor for cavitation.
Soft clay	Not recommended for jet suction models. For Models 3 and 4 reduce suction speed and monitor for blockage regularly.
Stiff clay	Not suitable for jet suction operation. Route around or use excavator to break up before dredging.
Production rate drops	Identify cause: (1) Check pump pressure — low pressure indicates suction blockage or impeller wear. (2) Check discharge pipe — blockage in discharge reduces output. (3) Check engine RPM — confirm engine is at rated speed. Resolve the identified cause before resuming full production.
Blocked suction pipe	Stop pump. Raise suction arm from the obstruction using manual winch. Lower back to dredging position and restart pump.
Blocked discharge pipe	Stop pump. Loosen pipe connections one at a time starting from the pump outlet to locate and clear the blockage. Flush with clean water before restarting.

6.3 Winch & Anchor Operating Guidelines

1	Always set all 4 anchors before commencing dredging swing. Confirm each anchor is holding before releasing to tension.

2	If anchor line goes slack during dredging: immediately adjust brake tension on the winch drum to restore tension. Do not allow the vessel to drift uncontrolled.
3	To reposition vessel: raise the suction arm to travel position first. Release anchors in reverse order. Reposition vessel. Reset anchors before resuming dredging.
4	Inspect all anchor wires at the start of every shift. Never operate with wires that are kinked, frayed, or showing broken strands.
5	No spuds are fitted to these models — all positioning relies on the 4-anchor winch system.

7. Safety & Emergency Procedures

WARNING: In any emergency: STOP operations immediately, ensure crew safety first, then address the vessel or equipment.

7.1 Personal Protective Equipment (PPE)

The following PPE is mandatory for all personnel on board at all times during operations:

TASK	TASK / SITUATION — MANDATORY PPE
All personnel on deck	Life jacket (worn at all times on water), hard hat, steel-toe boots, hi-visibility vest
Engine maintenance	Life jacket, hard hat, safety glasses, heat-resistant gloves, safety boots
Pump and pipe maintenance	Life jacket, hard hat, safety glasses, chemical-resistant gloves, safety boots
Cutter / suction head inspection	Life jacket, hard hat, safety glasses, cut-resistant gloves, safety boots
Fuelling operations	No smoking within 5 metres. Life jacket, chemical-resistant gloves, safety glasses. Ground static before fuelling.
Working at height on ladder structure	Life jacket, hard hat, full-body harness clipped to vessel structure, safety boots
Working in bilge / confined space	Life jacket, gas detector, buddy system — never enter bilge alone

7.2 Fire Emergency Procedure

1	Shout 'FIRE' to alert all crew. Activate emergency stop to shut down engine and electrical systems.
2	Locate the nearest fire extinguisher (positioned at bow and stern on each vessel). Use CO_2 extinguisher for engine/electrical fires. Dry powder for fuel fires.
3	If fire cannot be controlled within 60 seconds: call for evacuation of all crew.
4	All crew don life jackets and proceed to the designated muster point.
5	If vessel is near shore, move to shore immediately. If offshore, deploy life rings and await rescue.
6	Contact emergency services and company management as soon as crew are safe.

7.3 Flooding / Sinking Risk Procedure

1	If flooding is detected in the hull: immediately shut down engine and all electrical systems to prevent electrical hazard.
2	All crew don life jackets immediately.
3	Attempt to identify and block the source of flooding if it can be done safely within 2 minutes.
4	If flooding cannot be controlled: evacuate all crew to shore or nearest safe vessel.
5	Contact company management and port authority / coastguard immediately.

7.4 Man Overboard Procedure

1	Shout 'MAN OVERBOARD' to alert all crew. Note the position and time.
2	Throw the life ring (located at the head of the dredger) to the person in the water. Do not jump in after them unless you are a trained rescue swimmer.
3	Deploy diver for recovery if life ring alone is not sufficient and conditions are safe for diving.
4	Stop all engine and propulsion to prevent propeller or suction hazard to the person in the water.
5	Once person is recovered: check for injuries, administer first aid, and contact emergency services.
6	Record the incident and report to company management immediately.

7.5 Engine / Mechanical Failure at Sea

1	If engine fails: immediately raise suction arm to travel position using manual winch to prevent damage.
2	Deploy all 4 anchors to hold vessel position while fault is investigated.
3	Troubleshoot by tracing the fault — check fuel, oil pressure, coolant temperature, and battery before calling for assistance.
4	If pump fails suddenly: stop engine, flush pump with any available clean water, then investigate seal condition and impeller.
5	If fault cannot be resolved by the crew: contact company technical support and arrange tow or assistance from shore.
6	Prepare vessel for tow: remove all lines from water, secure all loose equipment, confirm towline attachment point at bow.

7.6 Fuel / Oil Spill Procedure

1	Stop fuelling / leaking source immediately. Shut off relevant valves.
2	Contain the spill using absorbent materials (rags, absorbent pads) stored on the vessel.
3	Do not allow spilled fuel or oil to enter the water. Use oil boom if available.
4	Report any spill to water to company management, port authority, and the relevant environmental agency immediately.
5	Record the incident in the vessel log.

8. Lubrication Chart & Fluid Specifications

TASK	FLUID / LUBRICANT SPECIFICATION
Engine oil	Total Ruby X — confirm correct viscosity grade with engine manual for ambient temperature
Cutter / suction bearing grease	Confirm grade with supplier — apply weekly and at each 200-hour service
Hydraulic fluid	Confirm grade with supplier — change at 200-hour intervals or when contaminated
Gearbox oil	Confirm grade and capacity with gearbox manufacturer — change every 200 hours / quarterly
Total grease points on vessel	7 points — grease all 7 points at every 200-hour service and as required
Recommended tools for service	Complete set of spanners (metric). No special tools required for routine maintenance.

NOTE: Always use the recommended lubricant grades. Using incorrect oil grades in the engine or gearbox will void the service warranty and may cause premature failure.

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